

Qualification Pack



Robot Systems Integration

QP Code: ASC/Q8318

Version: 1.0

NSQF Level: 4.5

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ASC/Q8318: Robot Systems Integration

Brief Job Description

The robot system integrator must provide technical solutions to the robotization of all or part of a system by incorporating a multi-articulating arm, together with the associated handling tools or special processes (such as handling, machining, painting, and welding), to increase competitiveness and supporting the ergonomics, health and safety of the users and people around them.

Personal Attributes

The person should be result oriented with good technical and analytical skills, should have Excellent Interpersonal Skills, communication and presentation skills and a good team player. They should have ability to manage projects, prioritizing of work and mentoring the budding engineers.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ASC/N8356: Work organization and management \(Robot Systems Integration\)](#)
2. [ASC/N9820: Communication and interpersonal skills \(Robot Systems Integration\)](#)
3. [ASC/N8357: Layout and design](#)
4. [ASC/N8358: Installation and connectivity](#)
5. [ASC/N8359: Automation and programming](#)
6. [ASC/N8360: Commissioning, maintenance, and troubleshooting](#)
7. [ASC/N8361: Documentation, briefing, and reporting](#)

Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Automotive Product Development
Country	India
NSQF Level	4.5

Qualification Pack

Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL
Minimum Educational Qualification & Experience	No formal education prescribed
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	25 Years
Last Reviewed On	NA
Next Review Date	08/02/2026
NSQC Approval Date	08/02/2024
Version	1.0
Reference code on NQR	QG-4.5-AU-01819-2024-V1-ASDC
NQR Version	1

Qualification Pack

ASC/N8356: Work organization and management (Robot Systems Integration)

Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising the use of resources.

Scope

The scope covers the following :

- Manage work and resources

Elements and Performance Criteria

Manage work and resources

To be competent, the user/individual on the job must be able to:

- PC1.** prepare and maintain a safe, tidy, and efficient work area
- PC2.** prepare self for the tasks in hand, including full regard to health, safety, and environment
- PC3.** schedule work to maximize efficiency and minimize disruption
- PC4.** select and use all equipment and materials safely and in compliance with manufacturers' instructions
- PC5.** apply or exceed the health and safety standards applying to the environment, equipment, and materials
- PC6.** restore the work area to an appropriate state and condition
- PC7.** contribute to team performance both broadly and specifically
- PC8.** give and take feedback and support

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** principles and methods of safe work execution
- KU2.** purposes, uses, care and maintenance of all equipment together with their safety implications
- KU3.** environmental and safety principles and applications with regard to good housekeeping in the work area
- KU4.** principles of effective communication
- KU5.** principles of effective collaboration
- KU6.** scope and limits of ones own and others roles, responsibilities, and duties, both individually and collectively
- KU7.** parameters within which activities must be planned
- KU8.** principles and techniques for time management

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Manage work and resources</i>	15	25	-	10
PC1. prepare and maintain a safe, tidy, and efficient work area	2	3	-	1
PC2. prepare self for the tasks in hand, including full regard to health, safety, and environment	2	3	-	2
PC3. schedule work to maximize efficiency and minimize disruption	2	3	-	1
PC4. select and use all equipment and materials safely and in compliance with manufacturers' instructions	2	4	-	2
PC5. apply or exceed the health and safety standards applying to the environment, equipment, and materials	2	3	-	1
PC6. restore the work area to an appropriate state and condition	1	3	-	1
PC7. contribute to team performance both broadly and specifically	2	3	-	1
PC8. give and take feedback and support	2	3	-	1
NOS Total	15	25	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8356
NOS Name	Work organization and management (Robot Systems Integration)
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Automotive Product Development
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Qualification Pack

ASC/N9820: Communication and interpersonal skills (Robot Systems Integration)

Description

This unit is about communicating with team members, superior and others effectively.

Scope

The scope covers the following :

- Communicate effectively with others

Elements and Performance Criteria

Communicate effectively with others

To be competent, the user/individual on the job must be able to:

- PC1.** interact with a range of business and industry, modelling professional conduct at all times
- PC2.** communicate by oral, written, and electronic means to ensure clarity, effectiveness, and efficiency
- PC3.** use a standard range of communication technologies
- PC4.** discuss complex technical principles and applications with others
- PC5.** use active listening and questioning techniques
- PC6.** read, interpret, and extract technical data and instructions from documentation in any available format
- PC7.** complete reports and respond to issues and questions arising
- PC8.** respond to clients' and personnel's needs face to face and indirectly
- PC9.** gather information and prepare documentation as required by the client and other individuals and groups

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational cultures and behaviours within business and industry
- KU2.** purposes and range of required documentation in paper and electronic forms
- KU3.** technical language associated with the occupation and sector
- KU4.** standards required for routine and exception reporting in oral, written, and electronic form
- KU5.** good practice in communication with clients, team members, and others
- KU6.** purposes and techniques for generating, maintaining, and presenting records for ones own and others uses

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate effectively with others</i>	15	25	-	10
PC1. interact with a range of business and industry, modelling professional conduct at all times	2	3	-	1
PC2. communicate by oral, written, and electronic means to ensure clarity, effectiveness, and efficiency	1	3	-	2
PC3. use a standard range of communication technologies	2	3	-	1
PC4. discuss complex technical principles and applications with others	2	3	-	1
PC5. use active listening and questioning techniques	2	3	-	1
PC6. read, interpret, and extract technical data and instructions from documentation in any available format	1	2	-	1
PC7. complete reports and respond to issues and questions arising	2	3	-	1
PC8. respond to clients' and personnel's needs face to face and indirectly	2	3	-	1
PC9. gather information and prepare documentation as required by the client and other individuals and groups	1	2	-	1
NOS Total	15	25	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9820
NOS Name	Communication and interpersonal skills (Robot Systems Integration)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Qualification Pack

ASC/N8357: Layout and design

Description

This NOS is about finalizing layout and design of the system.

Scope

The scope covers the following :

- Prepare layout and design of the system

Elements and Performance Criteria

Prepare layout and design of the system

To be competent, the user/individual on the job must be able to:

- PC1.** acquire and check instructions and guidance for given assignments
- PC2.** identify and resolve areas of uncertainty within the parameters of the brief
- PC3.** carry out initial systems design for given industrial applications
- PC4.** inspect installation sites or use alternative methods to test the applicability of initial systems design
- PC5.** optimize systems designs within the parameters of the given industrial applications
- PC6.** incorporate the dimensioning of electrical and pneumatic systems
- PC7.** determine the role of pneumatic engineering in the choice and connection of controls and activators
- PC8.** carry out systems analyses for risk assessment
- PC9.** itemize the requirements and implications of installation and integration in relation to: robots, ancillary equipment, and tools, human resources and time, estimated impacts on production during installation, estimated impacts on production following installation, operating parameters and risk management
- PC10.** present proposals for consideration and approval, and adjust as required

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the principles and relevant applications of computing and electronics
- KU2.** the relevant practical applications of engineering science and technology
- KU3.** the relevant practical implications of physical principles and interrelationships
- KU4.** the principles and relevant applications of electrical engineering and pneumatics
- KU5.** the design, uses, repair and maintenance needs of relevant machinery and tools
- KU6.** the principles and applications of robots, robotic tools and equipment mounted on robots and in robotic cells

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- KU7.** principles and methods of systems analysis to determine how conditions, operations and the environment will affect outcomes
- KU8.** principles and applications for incorporating and integrating robots within industrial systems, such as: payload settings, reach studies, motion optimization
- KU9.** principles of CAD and offline simulation tools used for layout and design of robot systems

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare layout and design of the system</i>	30	50	-	20
PC1. acquire and check instructions and guidance for given assignments	3	5	-	2
PC2. identify and resolve areas of uncertainty within the parameters of the brief	3	5	-	2
PC3. carry out initial systems design for given industrial applications	3	5	-	2
PC4. inspect installation sites or use alternative methods to test the applicability of initial systems design	3	5	-	2
PC5. optimize systems designs within the parameters of the given industrial applications	3	5	-	2
PC6. incorporate the dimensioning of electrical and pneumatic systems	3	5	-	2
PC7. determine the role of pneumatic engineering in the choice and connection of controls and activators	3	5	-	2
PC8. carry out systems analyses for risk assessment	3	5	-	2
PC9. itemize the requirements and implications of installation and integration in relation to: robots, ancillary equipment, and tools, human resources and time, estimated impacts on production during installation, estimated impacts on production following installation, operating parameters and risk management	3	5	-	2
PC10. present proposals for consideration and approval, and adjust as required	3	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8357
NOS Name	Layout and design
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Automotive Product Development
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

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ASC/N8358: Installation and connectivity

Description

This NOS is about installing and connecting the components of system.

Scope

The scope covers the following :

- Installation and connection of the system components

Elements and Performance Criteria

Installation and connection of the system components

To be competent, the user/individual on the job must be able to:

- PC1.** check that all items have been delivered according to specification, and follow up as required
- PC2.** organize the safe storage of all items, together with arrangements for their checking in and out
- PC3.** check that the pre-manufactured robot has been delivered ready to run, and follow up as required
- PC4.** connect robot system components according to instructions and documentation
- PC5.** assemble, position, and fix robotic tools and equipment according to instructions and documentation
- PC6.** align, fit, or assemble components, using hand tools, power tools, fixtures, or templates, according to specification
- PC7.** liaise with specialists for the correct electrical, pneumatic, and mechanical installation of robots and peripheral equipment
- PC8.** connect Input/Output (I/O) control signals between robot and peripheral equipment, either low voltage (24V) or Ethernet/Bus systems
- PC9.** perform tests during the installation process to ensure functionality
- PC10.** identify installation issues, consider alternative solutions, and implement selected solution(s) to resolve the issues
- PC11.** respect and take account of the receiving sites' requirements and characteristics, within the bounds of safe working, active risk management, and professionalism

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the norms and cultures of the receiving industrial sites
- KU2.** principles and methods for the safe receipt and ongoing management of equipment, tools, and materials
- KU3.** principles underlying the physical installation of robotics into production systems
- KU4.** principles and methods for assembling pre-manufactured robots in their positions for use

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- KU5.** principles and methods for assembling and fixing tools and equipment to the robots
- KU6.** principles underlying the positioning, connection and use of electrical power
- KU7.** principles underlying the positioning, connection and use of pneumatics
- KU8.** principles underlying the correct foundations and fixing methods required for installation of industrial robots and peripheral equipment

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Installation and connection of the system components</i>	30	50	-	20
PC1. check that all items have been delivered according to specification, and follow up as required	2	4	-	2
PC2. organize the safe storage of all items, together with arrangements for their checking in and out	2	3	-	1
PC3. check that the pre-manufactured robot has been delivered ready to run, and follow up as required	3	5	-	2
PC4. connect robot system components according to instructions and documentation	3	5	-	2
PC5. assemble, position, and fix robotic tools and equipment according to instructions and documentation	3	5	-	2
PC6. align, fit, or assemble components, using hand tools, power tools, fixtures, or templates, according to specification	3	5	-	2
PC7. liaise with specialists for the correct electrical, pneumatic, and mechanical installation of robots and peripheral equipment	3	5	-	2
PC8. connect Input/Output (I/O) control signals between robot and peripheral equipment, either low voltage (24V) or Ethernet/Bus systems	3	5	-	2
PC9. perform tests during the installation process to ensure functionality	3	5	-	2
PC10. identify installation issues, consider alternative solutions, and implement selected solution(s) to resolve the issues	3	5	-	2
PC11. respect and take account of the receiving sites' requirements and characteristics, within the bounds of safe working, active risk management, and professionalism	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8358
NOS Name	Installation and connectivity
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Automotive Product Development
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Qualification Pack

ASC/N8359: Automation and programming

Description

This NOS unit is about automation and programming of the system.

Scope

The scope covers the following :

- Automation and programming of the system

Elements and Performance Criteria

Automation and programming of system

To be competent, the user/individual on the job must be able to:

- PC1.** consult with client/personnel to clarify program intent
- PC2.** develop diagrams or flow charts of systems operations
- PC3.** write, analyse, review and rewrite programs, using flow charts and diagrams
- PC4.** create application software programs that are easy to document, understand and maintain
- PC5.** conduct trial runs of programs and software applications to ensure they will produce the desired robot and cell performance
- PC6.** write, update, and maintain computer programs or software packages to handle specific jobs
- PC7.** optimize robot motion performance and I/O handling to minimize cycle time/maximize throughput while retaining reliable operation
- PC8.** correct errors by making appropriate changes and rechecking the program to ensure that the desired results are produced
- PC9.** consult with other personnel to identify problems and suggest changes
- PC10.** implement new additional software and hardware options based on standard functionality
- PC11.** integrate simple and advanced sensors

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** computer capabilities and symbolic logic
- KU2.** principles governing the purposes and functions of computer hardware and software
- KU3.** principles and options for: manipulating robot coordinate frames, for robot, cell, and tooling, controlling robot motion, controlling robot input/output (I/O)
- KU4.** the principles, reasons or facts that provide the basis for breaking down information or data into separate parts
- KU5.** methods for obtaining information and data from all relevant sources
- KU6.** principles and methods for processing information and data
- KU7.** the software in use

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KU8. sensor integration: simple digital/electrical sensors, advanced sensors such as Vision or Force sensors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Automation and programming of system</i>	30	50	-	20
PC1. consult with client/personnel to clarify program intent	2	3	-	1
PC2. develop diagrams or flow charts of systems operations	2	4	-	2
PC3. write, analyse, review and rewrite programs, using flow charts and diagrams	3	5	-	2
PC4. create application software programs that are easy to document, understand and maintain	3	5	-	2
PC5. conduct trial runs of programs and software applications to ensure they will produce the desired robot and cell performance	3	5	-	2
PC6. write, update, and maintain computer programs or software packages to handle specific jobs	3	5	-	2
PC7. optimize robot motion performance and I/O handling to minimize cycle time/maximize throughput while retaining reliable operation	3	5	-	2
PC8. correct errors by making appropriate changes and rechecking the program to ensure that the desired results are produced	3	5	-	2
PC9. consult with other personnel to identify problems and suggest changes	3	5	-	2
PC10. implement new additional software and hardware options based on standard functionality	3	5	-	2
PC11. integrate simple and advanced sensors	2	3	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8359
NOS Name	Automation and programming
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Automotive Product Development
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Qualification Pack

ASC/N8360: Commissioning, maintenance, and troubleshooting

Description

This NOS is about commissioning, maintenance, and troubleshooting of the system

Scope

The scope covers the following :

- Commissioning, maintenance, and troubleshooting of the system

Elements and Performance Criteria

Commissioning, maintenance, and troubleshooting of the system

To be competent, the user/individual on the job must be able to:

- PC1.** investigate whether the robot and its peripheral equipment are responding to the programs' instructions
- PC2.** revise, repair or expand existing programs to increase operational efficiency or adapt to new requirements
- PC3.** repair or replace components as required
- PC4.** develop Human-Machine-Interface (HMI) applications for the users of the robot system, using HTML or other web technologies
- PC5.** advise on maintenance regimes to maximize efficiency and minimize disruption

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the formal requirements for successful site acceptance tests
- KU2.** the scope and limits of the technologies, methods operational environment
- KU3.** criteria and methods for testing equipment and systems
- KU4.** strategies for fault finding, problem solving and optimization
- KU5.** techniques and options for replacements and repairs
- KU6.** principles and techniques for generating creative and innovative solutions
- KU7.** principles and options for establishing and maintaining production maintenance regimes

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Commissioning, maintenance, and troubleshooting of the system</i>	15	25	-	10
PC1. investigate whether the robot and its peripheral equipment are responding to the programs' instructions	3	5	-	2
PC2. revise, repair or expand existing programs to increase operational efficiency or adapt to new requirements	3	5	-	2
PC3. repair or replace components as required	3	5	-	2
PC4. develop Human-Machine-Interface (HMI) applications for the users of the robot system, using HTML or other web technologies	3	5	-	2
PC5. advise on maintenance regimes to maximize efficiency and minimize disruption	3	5	-	2
NOS Total	15	25	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8360
NOS Name	Commissioning, maintenance, and troubleshooting
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Automotive Product Development
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Qualification Pack

ASC/N8361: Documentation, briefing, and reporting

Description

This NOS is about documentation and reporting of the completed work as per the organisation requirements.

Scope

The scope covers the following :

- Documentation, briefing, and reporting of work

Elements and Performance Criteria

Documentation, briefing, and reporting of work

To be competent, the user/individual on the job must be able to:

- PC1.** liaise with other personnel or departments for project integration
- PC2.** document design and development procedures according to requirements
- PC3.** compile and write documentation of program development and subsequent revisions, inserting comments in the coded instructions so that others can understand the computer programs
- PC4.** present and provide test results from the commissioning process
- PC5.** design or contribute to instructions and guidance to guide end users, with an emphasis on clarity and ease of use
- PC6.** provide the end user with a set of documentation in appropriate formats, including all necessary robot data such as: operating instructions, application specific fault messages - I/O Listings, user adjustable parameter (register) descriptions
- PC7.** review each part of the process of design, fabrication and assembly, and operation, against established criteria, including accuracy, consistency, time, and cost
- PC8.** contribute to individual and collective quality and contract review, responding to questions and challenges appropriately

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the role and importance of maintaining records of each stage of activity
- KU2.** the required media and formats of records and reports to ensure compliance with contracts, regulations and legislation, verification, and audit
- KU3.** the needs of users and specialists for information, guidance, and instructions in suitable forms (media, content, language, format, and presentation)
- KU4.** clients specific information needs
- KU5.** basic principles and techniques for briefing and training non-specialist end users
- KU6.** principles and techniques for critical review of own and others performance

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KU7. principles of common PC/Office software

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Documentation, briefing, and reporting of work</i>	15	25	-	10
PC1. liaise with other personnel or departments for project integration	2	3	-	1
PC2. document design and development procedures according to requirements	2	3	-	2
PC3. compile and write documentation of program development and subsequent revisions, inserting comments in the coded instructions so that others can understand the computer programs	2	3	-	1
PC4. present and provide test results from the commissioning process	2	4	-	2
PC5. design or contribute to instructions and guidance to guide end users, with an emphasis on clarity and ease of use	2	3	-	1
PC6. provide the end user with a set of documentation in appropriate formats, including all necessary robot data such as: operating instructions, application specific fault messages - I/O Listings, user adjustable parameter (register) descriptions	1	3	-	1
PC7. review each part of the process of design, fabrication and assembly, and operation, against established criteria, including accuracy, consistency, time, and cost	2	3	-	1
PC8. contribute to individual and collective quality and contract review, responding to questions and challenges appropriately	2	3	-	1
NOS Total	15	25	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	ASC/N8361
NOS Name	Documentation, briefing, and reporting
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Automotive Product Development
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

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6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N8356.Work organization and management (Robot Systems Integration)	15	25	0	10	50	5
ASC/N9820.Communication and interpersonal skills (Robot Systems Integration)	15	25	0	10	50	5
ASC/N8357.Layout and design	30	50	0	20	100	15
ASC/N8358.Installation and connectivity	30	50	0	20	100	15
ASC/N8359.Automation and programming	30	50	0	20	100	25
ASC/N8360.Commissioning, maintenance, and troubleshooting	15	25	0	10	50	25
ASC/N8361.Documentation, briefing, and reporting	15	25	0	10	50	10
Total	150	250	0	100	500	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.